Pengfei Zhang Applying for 2027 New Grads with F1 OPT (Expected Graduation Time: Dec2026, Mar2027, or Jun2027)
Email : **pengfz5@uci.edu** Github LinkedIn Personal Website

EDUCATION

Year	Degree	Institute	GPA
Fall 2022 -	PHD stu in Computer Science	University of California, Irvine	3.93/4.0
2017 - 2021	B.Eng in Computer Science	University of Science and Technology of China	3.45/4.3

RESEARCH INTEREST

Human Understanding, Multimodal Large Language Models, Multimodal Agentic RL

SELECTED PROJECTS

I. **VLM-based Multimodal Agents for Human-Centric Image/Video Understanding, Generation, and Editing**
This project focus on the understanding, generation, and editing of human-centric images or videos with complex motions, as essential personalized functions of Multimodal LLMs and Multimodal Agents.

Techniques: VLM-based Multimodal Agents Orchestration, MLLM Post-Training (DPO, GRPO, GDPO)

(e) **Identity Decay in Multi-Turn Human-Centric Image Editing: A Benchmark and a Multimodal Agentic Correction Framework** Under Review
—- Introduced a benchmark that measures the identity and fidelity decay for multi-turn Human-Centric Image Editing.
—- Proposed a correction pipeline in which a multimodal agent orchestrates the identity correction process using pretrained human segmentation and parsing models (sapiens for body, DML_CSR for face), and deterministic photoshop and image operators (such as tps warp, high-freq preservation transfer, etc), flattening the degradation curve.

(d) **KinMo: Kinematic-aware Human Motion Understanding and Generation** **ICCV 2025**
—- Designed a VLM-based dataset enhancement pipeline that extends the existing text-motion alignment dataset by incorporating proposed motion representation as body decomposition, enabling precise motion descriptions.
—- Introduced a hierarchical motion-text alignment approach that fuses motion details into text semantics, improving SOTA text-motion understanding and text-based motion generation models with more precise motion control.

(c) **LLM-Based Human-Agent Collaboration and Interaction Systems: A Survey** **ACL 2026**

(b) **Contextual Gesture: Co-Speech Gesture Video Generation through Context-aware Gesture Representation** **ACM MultiMedia (MM) 2025**
—- A framework that improves co-speech gesture video generation through chronological speech-gesture alignment that temporally connects two modalities, and a contextualized gesture tokenization that incorporate speech context into motion pattern representation through distillation.

(a) **Handformer2T: A Lightweight Regression-based model for Interacting Hands Pose Estimation from a single RGB Image** **WACV 2024**
—- A lightweight model which hand-level tokenization in the transformer based model for interacting hand pose estimation, where only one token was used for each hand.

II. **Speech Language Models and Multimodal Agents**
This project focus on the joint understanding of multimedia signals (texts, images, videos, speeches, audios, etc).

Techniques: MLLM Post-Training (DPO, GRPO, GDPO), Chain-of-thought (CoT) Distillation

(f) **AP-GRPO: Anchor-Gated Phonetic Alignment with Policy Optimization for Pathological Speech Reconstruction** **EMNLP 2026 (accepted)**
—- A GRPO framework with phonetic reward that aligns speech language models (SLMs) through audible-anchor preservation and inter-anchor phonetic compatibility to the original speech signal. AP-GRPO consists of: (i) an anchor-gated reward that matches anchors in clear regions; and (ii) an inter-anchor phonetic alignment reward that evaluates whether recovered contents are phonetically supported by the corresponding corrupted inter-anchor speech span.

(e) **DemMA: Dementia Multi-Turn Dialogue Agent with Expert-Guided Reasoning and Action Simulation.** **ACL 2026**
—- An expert-guided multi-turn dialogue agent designed to simulate dementia patients with clinically grounded personas by distilling expert-guided reasoning and action simulation.

(d) **MedSpeak: Knowledge Enhanced ASR Error Correction framework for Spoken Medical Question Answering.** **ICASSP 2026**

—- a knowledge-enhanced retrieval augmented generation (RAG) framework through a knowledge graph (KG) capturing semantic medical relationships and phonetic similarities between medical terms.

(c) AudioLens: Multi-Perspective Speech Clustering with Audio-Language Models *Under Review*

—- Introduce audio multi-perspective clustering, where a model directly partitions speech recordings according to a natural-language perspective while inferring both the number of clusters and their assignments. Construct AudioLens-Bench, a benchmark spanning multiple applications and evaluating cross-perspective generalization, and propose AudioLens -R1, a large audio-language model trained with reasoning distillation and direct preference optimization.

(b) **From Sounds to Scenes: A Benchmark for Evaluating Context-Aware Auditory Scene Understanding in Large Audio Language Models.** *Under Review*

—- A benchmark designed to evaluate whether Large Audio-Language Models can jointly interpret speech, events, and scene-level information to perform coherent semantic reasoning.

(a) Hallucination Rates of Speech Language Models on Pathological Speech Recognition: A Pilot Multi-Cohort Audit. **BSN 2025**

—- An evaluation of popular Speech Language Models (SLMs) on Pathological Speech Recognition at different hallucination types (insertion, fluent completion, entity fabrication, semantic substitution).

III. **Personalized LLM Agents on Healthcare**

This project aims to create personalized conversational healthcare agents, tailored to individual patients.

Techniques: Steering Vector for LLM Personalization, Retrieval Augmented Generation (RAG)

(c) **Can the Environment Speak for Itself? T^2-GRPO: Turn-Trajectory Group Relative Policy Optimization for Caregiver Agents** Under Review

—- introduces a turn–trajectory optimization scheme for long-horizon caregiver agents that derives dense turn-level rewards from patient-state transitions, separately normalizes them with trajectory-level outcomes via centered-rank group-relative advantages, and incorporates a binary safety veto to enforce hard behavioral constraints.

(b) **DEMENTIA-PLAN: An Agent-Based Framework for Multi-Knowledge Graph Retrieval Augmented Generation in Dementia Care.** **AAAI 2025**

—- Introduced an innovative retrieval-augmented generation framework that enhances conversations with mild-stage dementia patients by intelligently integrating multiple knowledge graphs with LLMs.

(a) **Adaptive Constraint Relaxation in Personalized Nutrition Recommendations: An LLM-Driven Knowledge Graph Retrieval Approach** **AMIA 2025 (Oral)**

—- Designed an LLM-Driven Knowledge Graph Retrieval System for Personalized Food Recommendation, to resolve "all-or-nothing" limitation for traditional recommendation system.

Vibe-Coding PROJECTS in Production

I. **Tiktok Ads - Chatgpt App**: *A conversational advertisement creation app on the ChatGPT App Store, that helps novice advertisers turn a product page URL into a live Tiktok Smart+ campaign.*
Techniques: MCP Layer, Controlled Video Generation, Agent Self-Reflection, Multi-Agent Orchestration

II. **Claworld**: *An Agent-to-Agent Platform for agents to interact with each other, trading skills or carrying on tasks.*
Techniques: Agent Society, Self-Evolving Agents

Working Experience

- Machine Learning Engineer Intern at Bytedance in San Jose *Mar. 2026 - Sep. 2026*
  —- Multimodal Agents Tool Usage and Social Intelligence of Agents (Publications I.e, Proj I, II).
- Applied Scientist Intern at Amazon Web Services in San Jose *Jun. 2025 - Sep. 2025*
  —- Multimodal LLMs recognition, understanding, and reasoning in low-resource languages (Publications II.a, II.d).
- Research Science Intern at Flawless. AI. Inc in Los Angeles *Jun. 2024 - Sep. 2024*
  —- Multimodal LLMs for Human Motion Understanding and Multimedia Generation/Editing (Publications I.c, I.b).

AWARDS

- Dean’s award from UCI *2022-2024*
- National Encouragement Scholarship (top 20%) from USTC *2018-2020*